Beregningsteknik for Computer Ingeniører

MM 4: Induction and recurrence

Topics:

- Principle of mathematical induction
- Examples of using induction
- Mistaken induction proofs
- Strong induction
- Recursive definitions and recursively defined functions.

Principle of mathematical induction

PRINCIPLE OF MATHEMATICAL INDUCTION To prove that $P(n)$ is true for all positive integers n, where $P(n)$ is a propositional function, we complete two steps:

BASIS STEP: We verify that $P(1)$ is true.

INDUCTIVE STEP: We show that the conditional statement $P(k) \to P(k+1)$ is true for all positive integers k.

Mathematical argument consisting of:

- **A proposition P(n)**: A particular statement/function that is asserted true for k ≥ b (or for all non-negative or positive integers)

- **A base case/basis step**: Asserting that P(b) is true for the initial value b
- **An inductive hypothesis:** Assume we know P(k) is true for *some* k ≥ b
- **An inductive step:** ”If P(k) is true, we show that P(k+1) is true

  (P(k) → P(k+1): if ”P(k) then P(k+1)”, or ”P(k) only if P(k+1)”)
- **Conclusion:** P(n) is true for all positive integers n ≥ b

Strong (complete) induction

- Example: Show that if n is an integer greater than 1, then n can be written as the product of primes.

- Strong induction: In the inductive step we assume (inductive hypothesis) that P(j) is true for all 1 <= j <= k (the basis step is the same)
  - **We show** that if P(j) is true for all 1 <= j <= k then P(k+1), k > 1

STRONG INDUCTION To prove that $P(n)$ is true for all positive integers n, where $P(n)$ is a propositional function, we complete two steps:

BASIS STEP: We verify that the proposition $P(1)$ is true.

INDUCTIVE STEP: We show that the conditional statement $[P(1) \wedge P(2) \wedge \cdots \wedge P(k)] \rightarrow P(k+1)$ is true for all positive integers k.

Recursive definitions

- Sometimes it is possible to define an object (function, sequence, set, algorithm, structure) in terms of itself. This process is called **recursion.**
- In some instances recursive definitions of objects may be much easier to write
- Can you think of a function that is defined recursively?

Recursively defined functions

To define a function on the set of nonnegative integers

1. Specify the value of the function at 0 (or any other initial value)

2. Give a rule for finding the function's value at n+1 in terms of the function's value at integers $i \leq n$.

- Well defined functions: the value of the function at this integer is determined in an unambiguous way.

- Example 1.

$$f(0) = 3,$$
$$f(n+1) = 2f(n) + 3.$$

- Example 2. Give a recursive definition of a^n, where a is a nonzero real number and n is a nonnegative integer.

Recursively defined algorithms

- **Definition**:
  - "An algorithm is called recursive if it solves a problem by reducing it to an instance of the same problem with smaller input."
  - Example:

    procedure $\boxed{\textit{power}}$(a: nonzero real number, n: nonnegative integer)
    if $n = 0$ **then return** 1
    else return $a \cdot \boxed{\textit{power}}(a, n-1)$
    {output is a^n}

    - Proof of correctness by induction:

      … if n = 0, then *power*(a,0) returns 1 – ok, since $a^0 = 1$

      P(k): *power*(a,k) = a^k, k >= 0 TRUE

      P(k) → P(k+1) … *power*(a,k+1) = $a \cdot$*power*(a,k) = $a \cdot a^k = a^{k+1}$
- The alternative to recursion is iteration – start from the base case and successively apply the recursive definition

Recursively defined sequences

- Fibonacci numbers

  F(0) = 0, F(1) = 1

  F(n+1) = F(n) + F(n-1) for n = 1, 2, ...

- Fibonacci example, with links to linear algebra

  Show that whenever $n \geq 3$, $f_n > \alpha^{n-2}$, where $\alpha = (1 + \sqrt{5})/2$.